

Lab2

SRS Standard (IEEE Standard 830)

1: Introduction

- (i) Purpose
- (ii) Scope:
- (iii) Definition/Acronym/Abbreviation
- (iv) References
- (v) Overview

2. General Description

This section elaborates the functionality of the intended software in finer details containing following sub-sections:

- (i) Product Perspective
- (ii) Product Functions:
- (iii) User Characteristics:
- (iv) General Constraints:
- (v) Assumptions and dependencies:

3. Specific Requirements:

- (i) External interface requirements
- (ii) Functional requirements
- (iii) Performance requirements
- (iv) Design constraints :
- (v) Logical database requirement
- (vi) Software system attributes

Reliability

Availability

Security

Maintainability explains

4. Appendixes Section

Library Management System

Introduction:

- (i) **Purpose:** The purpose of this project is to provide a good arrangement or idle system in library management like arrangement of book, employee details, and issue in book availability etc.
- (ii) **Scope:** Library management system update the old file based system to online system.

The project is specifically designed for the use of librarians and library users.

Easily implemented under rush situations.

(iii) Definitions, Acronyms and Abbreviations:

JAVA or Dot Net

SQL- Structured query Language

DFD – Data Flow Diagram

ER- Entity Relationship

UML - Unified Modeling Language

IDE- Integrated Development Environment

SRS- Software Requirement Specification

ISBN - International Standard Book Number

(iv) References:

Book:

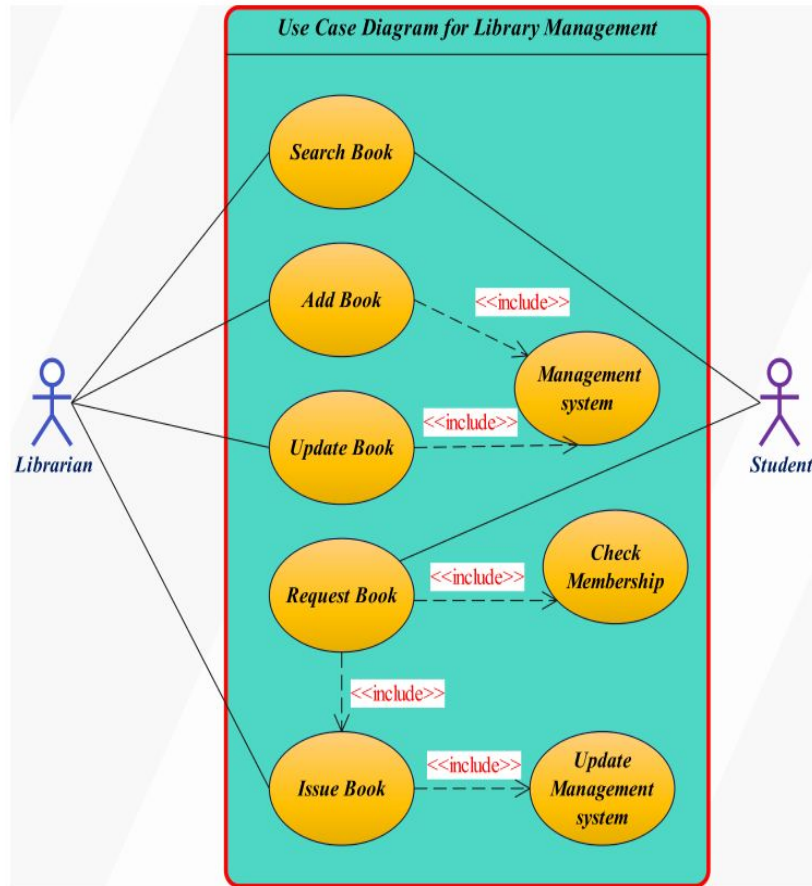
Website:

- (v) **Overview:** In the library management system we are defining various component in different section, in section one we are showing introduction, purpose, definition and references.
- In Section two we are discussing project perspective, project function, general constrain and Assumption and dependencies. And in section three we are discussing specific requirements of project
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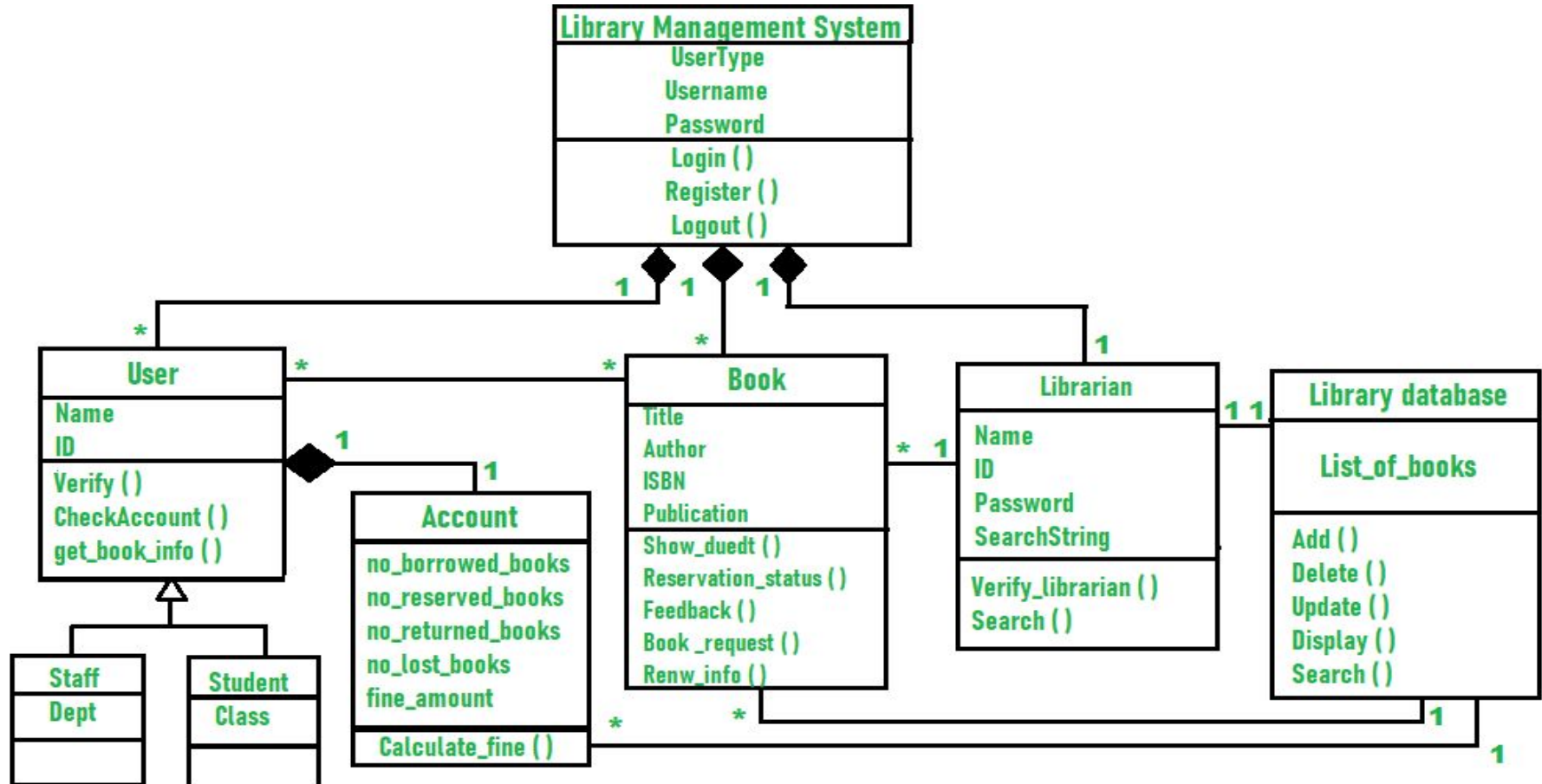
General Description:



(i) Product Prospective: Use Case Diagram



Product Function



CLASS DIAGRAM FOR LIBRARY MANAGEMENT SYSTEM

(iii) User Characteristics:

Features for the Librarian are:-

- A librarian can issue a book to the member.
- Can view the different categories of books available in the Library
- Can view the List of books available in each category
- Can take the book returned from students
- Add books and their information to the database
- Edit the information of existing books
- Can check the report of the existing books
- Can check the report of the issued books
- Can access all the accounts of the students

Features that are available for the Members/Students are:-

- Can view the different categories of books available in the Library
- Can view the List of books available in each category
- Can own an account in the library.
- Can view the books issued to him
- Can put a request for a new book
- Can view the history of books issued to him previously
- Can search for a particular book

General Constraints: This system will run in windows environment. The hardware requirement of this project is; Minimum 80GB hard disk, 1 GB RAMS, and works with common peripheral device like mouse, printer, and keyboard.

(v) **Assumptions and dependencies:**

The assumptions are:-

- ❖ The coding should be error free
- ❖ The system should be user-friendly so that it is easy to use for the users
- ❖ The information of all users, books and libraries must be stored in a database that is accessible by the website
- ❖ The system should have more storage capacity and provide fast access to the database
- ❖ The system should provide search facility and support quick Transactions
- ❖ The Library System is running 24 hours a day

Specific Requirements:

(i) External interface requirements:

- ❖ This Software provides good GUI based interface for user and administrator.
- ❖ Provide a login page where user and admin both have access.
- ❖ Librarian has separate login access of control panel where he can add or remove book availability or do other works.

(ii) Functional requirements: Here Librarian manage all database and check all necessities like payment detail, user detail and book availability etc.

(iii) **Performance Requirement:**

- The performance of the system should be fast and accurate
- The system should be able to handle large amount of data. Thus it should accommodate high number of books and users without any fault.

(iv) **Design constraints:** There are number of factor in environment to create barrier in system development. Earlier we faced issue in hardware then we increase the capacity of system.

(v) **Logical database requirement:** For secure database we are using SQL and and use RDBMS functionality in database

(vi) **Software system attributes:**

Reliability: Provide a secure and reliable system.

Availability: Our system will be work anytime and anywhere. Librarian can access and maintain the database in home also.

Security: Provide a good and secure system use used SSL certificate in our webpage.

Maintainability: We will provide a dedicated server and dedicated team for service.

Appendixes:

Glossary:

- Administrator: A login id representing a user with user administration privileges to the software
- User: A general login id assigned to most users
- Client: Intended users for the software
- SQL: Structured Query Language; used to retrieve information from a database
- SQL Server: A server used to store data in an organized format
- Layer: Represents a section of the project